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VARIABLE POWER SUPPLY
For Up To 3A Current

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The circuit presented here can provide up to three amperes of current for a 5V, 6V, 8V, or 9V supply. It uses the popular adjustable voltage regulator LM317, which is inexpensive and is used generally for variable power supplies with up to 1.5 ampere current.

With the addition of PNP transistor TIP127 in the circuit, we get a 5V to 9V variable power supply with 3A current rating. However, suitable heatsinks for LM317, TIP127, LM7809 and an exhaust fan for cooling the TIP127 are required.

The author's prototype is shown in Fig. 1. The block diagram of the variable power supply is shown in Fig. 2 and its circuit diagram in Fig. 3.

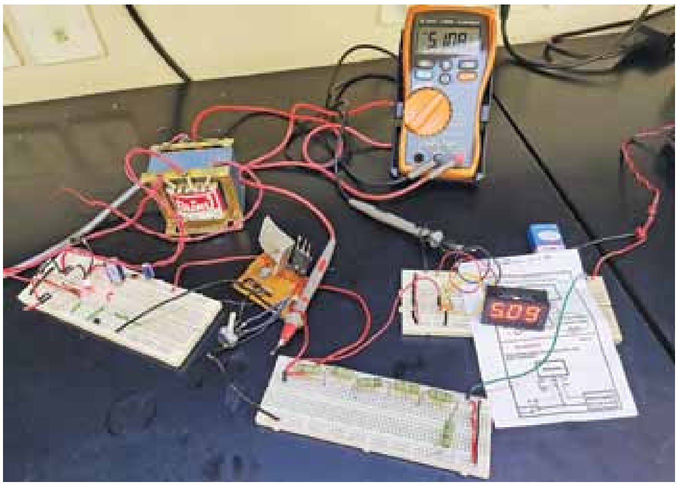

Fig. 1: Author's prototype

Circuit and working

The variable power supply is built around step-down transformer X1, four 1N5408 rectifier diodes D1 through D4, an LED, TIP127 transistor T1, LM317 adjustable voltage regulator IC1, 9V voltage regulator IC2, and a DC digital panel meter (DPM). A 22-ohm, 2-watt resistor (R1) is connected between the base and emitter of T1 to control the current. Pot VR1 is used to control the output voltage. The output voltage can be seen on the DPM.

The output voltage can be set to 9V and the current drawn is around 3A. The voltage across resistor R1 makes transistor T1 to conduct and most of the current is drawn from

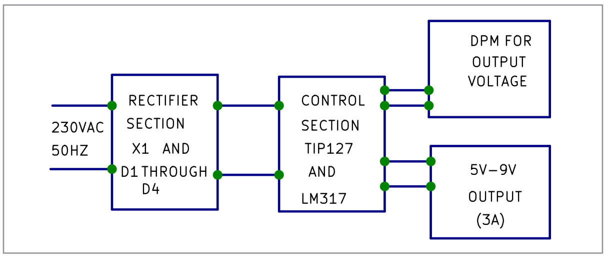

Fig. 2: Block diagram

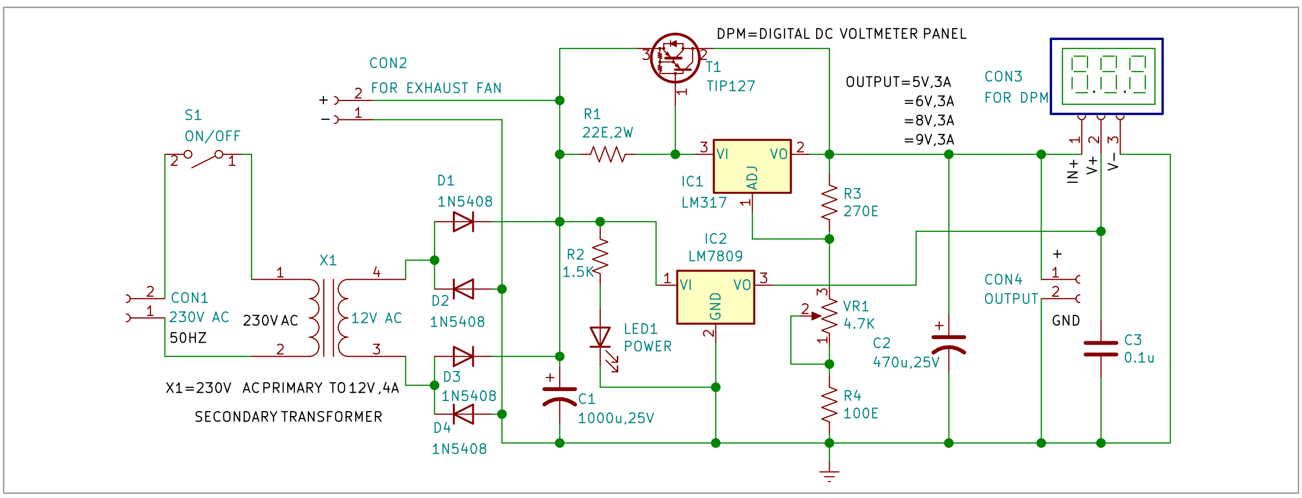

Fig. 3: Circuit diagram